Lecture MMNS 21.05.2026

Van der Pol equation

$q = q(t) \quad \ddot{q} + q - \varepsilon(1 - q^2)\dot{q} = 0, \ \varepsilon > 0$

$t \in \mathbb{R}$

$\rightsquigarrow x = (x_1, x_2) \rightsquigarrow \dot{x} = v(x) = \begin{pmatrix} x_2 \\ -x_1 + \varepsilon(1 - x_1^2)x_2 \end{pmatrix}$

$\rightsquigarrow (0,0)$ equi

$Dv(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & \varepsilon \end{pmatrix}$ $\quad \det D > 0$

$\det = 1 > 0$

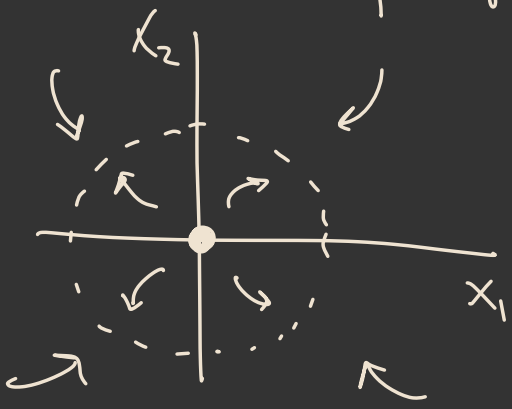
$\lambda_{1,2} = \frac{\varepsilon}{2} \pm \frac{\sqrt{\varepsilon^2 - 4}}{2} \Rightarrow$ * $\varepsilon < 2 \Rightarrow$ unstable spiral

* $\varepsilon > 2 \Rightarrow$ unstable node

$\rightsquigarrow \frac{d}{dt} H(x) = \frac{d}{dt}\left(\frac{1}{2}(x_1^2 + x_2^2)\right)$

$= x_1\dot{x}_1 + x_2\dot{x}_2 = \varepsilon(1 - x_1^2)x_2^2$

$\frac{d}{dt} H(x)$: $\begin{cases} > 0, & |x_1| < 1 \Rightarrow \text{giving energy to the system} \\ < 0, & |x_1| > 1 \Rightarrow \text{dissipating} \end{cases}$



STEP 1 : Heuristics

$(x_1, x_2) \longmapsto (E, \theta)$, $x_1 = \sqrt{2E}\cos\theta$

$x_2 = \sqrt{2E}\sin\theta$

$$\begin{cases} \dot{E} = \varepsilon\left(1 - 2E\cos^2\theta\right) 2E\sin^2\theta =: \varepsilon F(E,\theta) \\ \dot{\theta} = -1 + \varepsilon \sin\theta\cos\theta\left(1 - 2E\cos^2\theta\right) \end{cases}$$

slow-fast dynamics (perturbation theory)

$$\begin{cases} \dot{E} = \varepsilon \bar{F}(E) \\ \dot{\theta} = -1 + \varepsilon \sin\theta\cos\theta\,(1 - 2E\cos^2\theta) \end{cases}$$

$$\bar{F}(E) = \frac{1}{2\pi}\int_0^{2\pi} F(E,\theta)\,d\theta \overset{[\dots]}{=} E\left(1-\frac{E}{2}\right)$$

$$\bar{F}(E) = 0, \quad E_1 = 0, \quad E_2 = 2$$

$\rightsquigarrow$ $\exists$ limit cycle $r = 2$

$\varepsilon = 0$



$\varepsilon \neq 0$



STEP 2 : make step 1 rigorous.

$$1 \ll \Delta t \ll \frac{1}{\varepsilon} \qquad (\varepsilon \ll 1)$$

$$\begin{cases} \dot{E} = \varepsilon F(E,\theta) = \varepsilon\, 2E\sin^2\theta\,(1 - 2E\cos^2\theta) \\ \dot{\theta} = -1 + \varepsilon G(E,\theta) = -1 + \varepsilon \sin\theta\cos\theta\,(1 - 2E\cos^2\theta) \end{cases}$$

Since $\dot{E} \approx \varepsilon \ll 1$

$$\dot{E} \approx \frac{\Delta E}{\Delta t} \qquad t < \sigma < t+\Delta t$$

$$E(\sigma) \simeq E(t) + \mathcal{O}(\varepsilon \Delta t)$$

$\dot{E} \sim \varepsilon$

$$\theta(\sigma) \simeq \theta(t) + (-1)(\sigma - t) + \mathcal{O}(\varepsilon \Delta t)$$

$$= \theta(t) + t - \sigma + \mathcal{O}(\varepsilon \Delta t)$$

$$\dot{E}(t) \simeq \frac{\Delta E}{\Delta t} = \frac{1}{\Delta t} \cdot \varepsilon \int_t^{t+\Delta t} F(E(\sigma), \theta(\sigma))\, d\sigma$$

$$= \frac{\varepsilon}{\Delta t} \int_t^{t+\Delta t} d\sigma \left[F\left(E(\sigma), \theta(t) + t - \sigma \right) + \mathcal{O}(\varepsilon \Delta t) \right]$$

$E(\sigma) \to E(t)$

$$= \frac{\varepsilon}{\Delta t} \int_t^{t+\Delta t} F(E(\sigma), -\theta(t) - t + \sigma)\, d\sigma + \mathcal{O}(\varepsilon^2 \Delta t)$$

$$= \frac{\varepsilon}{\Delta t} \int_{\alpha_1}^{\alpha_2} F(E(t), \alpha)\, d\alpha + \mathcal{O}(\varepsilon^2 \Delta t) + \frac{\varepsilon}{\Delta t}\, \mathcal{O}(\varepsilon \Delta t^2)$$

$\alpha = \sigma - t - \theta(t)$

$d\alpha = d\sigma$

$\alpha_1 = -\theta(t)$

$\alpha_2 = \Delta t - \theta(t) \overset{E(t) \to E(t+\Delta t)}{=} t + \Delta t - t - \theta(t)$

$$= \frac{\varepsilon}{\Delta t} \cdot \frac{\Delta t}{2\pi} \int_0^{2\pi} F(E(t),\alpha)\, d\alpha + \\ + \mathcal{O}(\varepsilon^2 \Delta t) + \mathcal{O}\left(\frac{\varepsilon}{\Delta t^2}\right)$$

$$\int_{\alpha_1}^{\alpha_2} \rightsquigarrow \int_0^{2\pi}$$

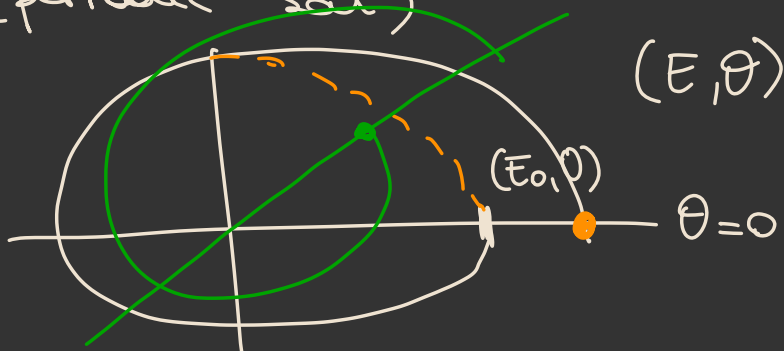
(α_1, α_2)

$$= \frac{\varepsilon}{2\pi} \int_0^{2\pi} F(E(t),\alpha)\, d\alpha + \varepsilon \left[\theta\left(\frac{1}{\Delta t^2}\right) + \varepsilon \mathcal{O}(\Delta t) \right]$$

$$= \varepsilon \bar{F}(E)$$

$$\Rightarrow \dot{E} \sim \frac{\varepsilon}{2\pi} \int_0^{2\pi} F(E,\theta)$$

STEP 3: $\exists$ limit cycle (by Poincaré section)
(periodic sol)



Let $t \longmapsto (E(t), \theta(t)) = \phi^t(E_0, \theta_0)$

sol of $\begin{cases} \dot{E} = \varepsilon F(E,\theta) \\ \dot{\theta} = -1 + \varepsilon G(E,\theta) \end{cases}$

with $\begin{cases} E(0) = E_0 \\ \theta(0) = \theta_0 \end{cases}$

We observe that

* $\dot{E} = 2\varepsilon E \sin^2\theta(1 - 2\varepsilon\cos^2\theta) \leq 2\varepsilon E$

$\Rightarrow E(t) \leq E_0 e^{2\varepsilon t} < +\infty$

$\varepsilon \ll 1$

$t < +\infty$

$\Rightarrow$ the energy does not blow up in finite time

$\Rightarrow$ the sol is global

* $|\dot{\theta} + 1| \leq \varepsilon(1 + 2E) \leq \varepsilon(1 + 2E_0 e^{2\varepsilon t})$

$\Rightarrow |\theta(t) + t| \leq \varepsilon(1 + 2E_0 e^{2\varepsilon t})t$

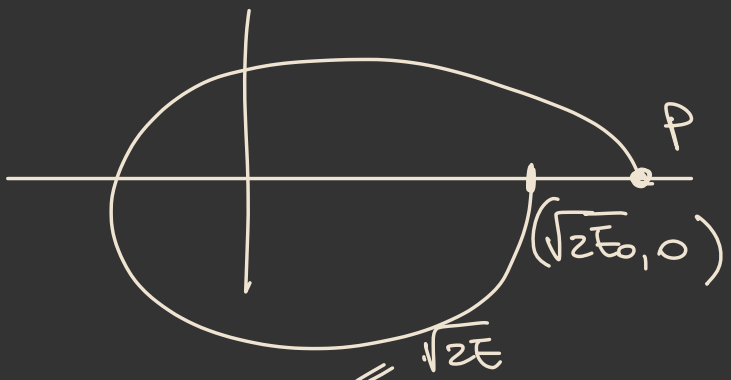
$\theta(0) = 0$

$\Rightarrow \theta(t) \approx t(1 + \varepsilon)$

To conclude

$$\begin{cases} E(t) \text{ does blow up} \\ \qquad (\text{does not increase too much}) \\ \theta(t) \sim t \end{cases}$$

$\Rightarrow \theta(t)$ is a 2π-periodic function



Therefore, we fix $r > 2$

$\exists\, \varepsilon_r > 0 : \forall\, \varepsilon < \varepsilon_r$ and $\forall\, t \in (0, 3\pi]$ s.t.

$$E_0 e^{2\varepsilon_r 3\pi} < 2 \quad \text{and} \quad \varepsilon_r (1+4r) < \frac{1}{4}.$$

With this choice, we can prove that

$\varepsilon(t) \le 2r$, $|\dot\theta(t)+1| < \varepsilon_r(1+\delta r)$

$\forall t \in (0, 3\pi]$, $\forall\ \varepsilon_0 \in (0, r]$, $\forall\ \varepsilon \in (0, \varepsilon_r)$

$$|\dot\theta(t)+1| < \frac{1}{4}$$

i.e.

$$\begin{cases} \dot\theta(t) < \frac{1}{4} - 1 = -\frac{3}{4} \\ \dot\theta(t) > -\frac{1}{4} - 1 = -\frac{5}{4} \end{cases}$$

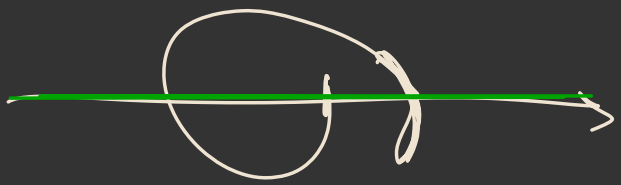
$$\Rightarrow \int_0^{3\pi} dt + \theta_0 = 0$$

$$\theta(3\pi) < -\frac{3}{4} \cdot 3\pi = -\frac{9\pi}{4}$$

$$\theta(3\pi) > -\frac{15}{4}\pi$$

$$(*) = -\frac{15}{4}\pi < \theta(3\pi) < -\frac{9\pi}{4} = -2\pi - \frac{\pi}{4}$$

$$-3\pi - \frac{3}{4}\pi$$



Therefore $t \longmapsto \theta(t)$ is monotone for $t \in [0, 3\pi]$

and (*) shows

$$\forall\, \varepsilon \in (0, \varepsilon_r),\ \forall\, (E_0, \theta_0) \in (0, r] \times \mathbb{R}$$

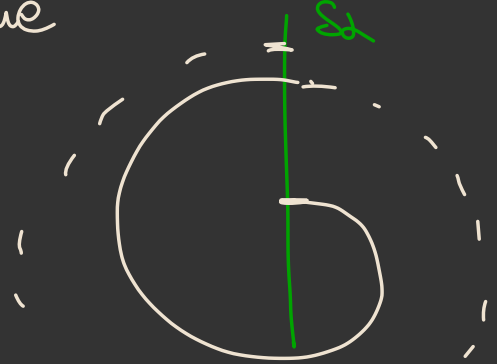
$$\exists\,!\ T_\varepsilon = T_\varepsilon(E_0, \theta_0) \in (0, 3\pi) \text{ s.t.}$$

$$\theta(T_\varepsilon) = \theta_0 - 2\pi$$

Therefore if $\varepsilon \in (0, \varepsilon_r)$, every segment $\alpha \in (0, 2\pi)$ S_α s.t.

$$S_\alpha := \{(E, \alpha) : E \in (0, r)\}$$

is a trasversal of the vector field where the trajectory is coming back for the first time



Let's consider $\alpha = 0$, we define $\varepsilon < \varepsilon_r$

$g_\varepsilon : (0, r) \longmapsto \mathbb{R}_+$ s.t.

$$g_\varepsilon(E_0) := E_0 + \int_0^{T_\varepsilon} \varepsilon F(E(s), \theta(s))\, ds$$

with $\left(E(t), \theta(t)\right) = \phi^t(E_0, 0)$

$$T_\varepsilon = T_\varepsilon(E_0, 0)$$

It remains to prove that $\exists$ isolated fixed points for g_ε

Tool: Implicit function thm.

From the eq. of $(E(t), \theta(t))$, we have

$$|E(t) - E_0| \le \int_0^t \varepsilon |F(E(s), \theta(s))|\, ds$$

$$\le 2\varepsilon t E_0 e^{2\varepsilon t}\left(1 + 2E_0 e^{2\varepsilon t}\right)$$

and

$$|\theta(t) + t| \le \varepsilon \int_0^t |G(E(s), \theta(s))|\, ds$$

$$\le 2$$

$$\leq \varepsilon t\left(1+2E_0 e^{2\varepsilon t}\right)$$

since $T_\varepsilon \in (0,3\pi)$, $\exists$ positive constant $C_0 = C_0(r)$ s.t. $\forall\ \varepsilon \in (0,\varepsilon_r)$

$$\begin{cases} \sup_{t\in[0,T_\varepsilon]} |E(t)-E_0| \leq \varepsilon C_0 \\ \sup_{t\in[0,T_\varepsilon]} |\theta(t)+t| \leq \varepsilon C_0 \\ |T_\varepsilon - 2\pi| \leq C_0\cdot\varepsilon \quad \text{because } \theta(T_\varepsilon) = -2\pi. \end{cases}$$

Remember that

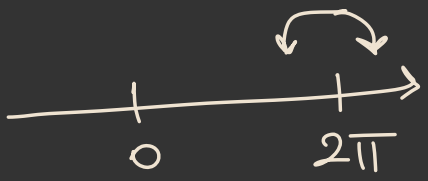
$$\int_0^{2\pi} F(E_0,-s)\,ds = [\dots] = 2\pi E_0\left(1-\frac{E_0}{2}\right)$$

then, we can estimate

$$\int_0^{T_\varepsilon} F(E(\sigma),\theta(\sigma))\,d\sigma = 2\pi E_0\left(1-\frac{E_0}{2}\right)+\varepsilon h$$

$$h = h(E_0, \varepsilon) := \int_{2\pi}^{\tau_\varepsilon} \frac{1}{\varepsilon} ds\, F(e_0, s) +$$

$$+ \int_0^{\tau_\varepsilon} \frac{ds}{\varepsilon} \left(F(E(s), \theta(s)) - \bar{F}(E_0, s) \right)$$



by regularity of F (by boundedness)

$\exists\ C_1 = C_1(r) > 0$ s.t.

$$\sup_{\varepsilon \in (0, \varepsilon_r)} \sup_{E_0 \in (0, r)} |h(E_0, \varepsilon)| \le C_1(r)$$

and it is possible to prove that

$\exists\ C_2 = C_2(r) > 0$ s.t.

$$\sup_{\varepsilon \in (0, \varepsilon_r)} \sup_{E_0 \in (0, r)} \left| \frac{\partial h(E_0, \varepsilon)}{\partial E_0} \right| \le C_2(r) \leftarrow$$

Therefore, we have

$$g_\varepsilon(E_0) = E_0 + 2\pi E_0\left(1 - \frac{E_0}{2}\right) + \varepsilon^2 h(E_0, \varepsilon)$$

with h satisfying the inequalities above